

Review-1: Progress Report

Topic

Reliability-Conditioned Prediction of Herb-Drug Interactions using Heterogeneous Graph Neural Networks The objective of this project is to build an AI-powered system that predicts potential adverse interactions between traditional herbs and pharmaceutical drugs. Unlike standard prediction models, our system computes a "Reliability Score" for its predictions by evaluating the quality, recency, and corroboration of the underlying evidence in the knowledge graph.

Data

We are utilizing the global gold-standard pharmaceutical database for this project:

- **Primary Dataset:** DrugBank XML Database (downloaded via Kaggle API `sergeguillemart/drugbank`).
- **Scale:** The raw dataset is 1.5GB in size.
- **Entities Extracted:** Successfully parsed over 5,000 distinct drugs/compounds and their structured metadata.
- **Edges/Interactions:** Extracted over 450,000 ground-truth interaction pairs along with textual clinical descriptions (evidence spans).
- **Phase 2 Expansion (Planned):** For the final research paper, the graph will be fused with traditional medicine datasets (IMPPAT and SymMap) to explicitly map Ayurvedic and Chinese herbal structures.

Finalise Flow Pipeline and GANTT Chart

System Flow Pipeline

1. **Data Ingestion:** Large XML corpus downloaded from Kaggle is parsed iteratively.
2. **Graph Construction:** Entities (Drugs/Herbs) are mapped to indices, and interactions are mapped to edge pairs to construct a PyTorch Geometric `HeteroData` graph.
3. **Cross-Attention GNN:** The `HeterogeneousGNNEncoder` learns node embeddings by passing messages across the graph, utilizing a multi-head cross-attention mechanism to weigh evidence reliability (corroboration, temporal recency, biomedical quality).
4. **Prediction:** The `LinkPredictor` outputs a probability of interaction and an explainable risk breakdown.
5. **User Interface:** A Streamlit-based web dashboard queries a FastAPI backend to surface predictions and real clinical evidence directly from the parsed graph.

GANTT Chart / Timeline

- **Week 1-2:** Literature review, architectural design, and environment setup.
- **Week 3-4 (Current):** Data acquisition (Kaggle DrugBank), iterative XML parsing, and data preprocessing into Graph JSON structures.
- **Week 5-6:** Implementation of the PyTorch Geometric Heterogeneous GNN and Link Predictor. Model ablation studies.
- **Week 7-8:** Integration of IMPPAT/SymMap for the herbal expansion. Model fine-

tuning.

- **Week 9-10:** Frontend dashboard finalization, evaluation metrics, and research paper drafting.

Any Other Progress (Till Data Preprocessing)

- **XML Parsing Engine:** Built a memory-efficient `xml.etree.ElementTree.iterparse` script that processes the 1.5GB DrugBank file without crashing the system memory.
- **Graph Mapping:** Successfully converted the raw XML nodes into a structured `drugbank_parsed.json` file.
- **Web Dashboard:** Developed and deployed a "pro-level" Glassmorphism UI (FastAPI + Streamlit) that currently queries the real preprocessed Kaggle data to demonstrate end-to-end connectivity.
- **Dependency Management:** Migrated the entire environment to `uv` for lightning-fast, deterministic package resolution.